Lab3 competition ROCstars

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Instructions

- Run all cells up to test evaluation section
- Set the random seed
- Run remaining cells

Libraries

```
library(e1071)
library(ggplot2)
```

```
Attaching package: 'ggplot2'

The following object is masked from 'package:e1071':

    element
```

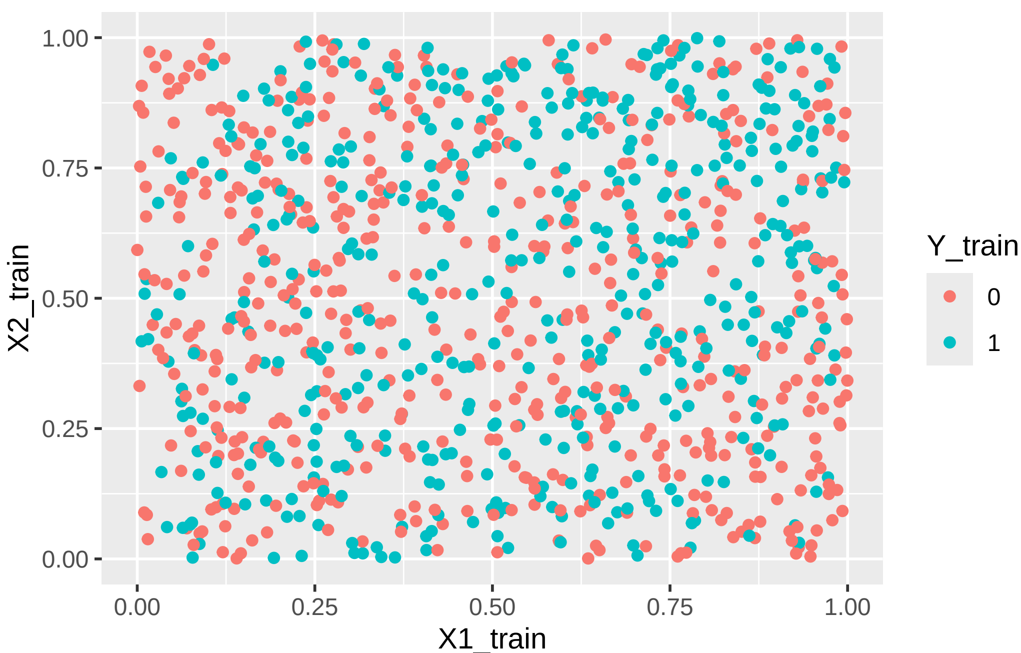
Train generation

```
generate_y <- function(x1,x2) { #two input parameters to generate the output y
  logit <- x1 -2*x2 -2*x1^2 + x2^2 + 3*x1*x2
  p <- exp(logit)/(1+exp(logit)) #apply the inverse logit function
  y <- rbinom(1,1,p) #y becomes a 0 (with prob 1-p) or a 1 with probability p
}
# Generate a training dataset with 1000 points
```

```r
set.seed(321)
n = 1000
X1_train <- runif(n,0,1)
X2_train <- runif(n,0,1)

#I'm going to use a for loop to generate 1000 y's
Y_train <- rep(0,n) #initializing my Y to be a vector of 0's
for (i in 1:n) {
  Y_train[i] <- generate_y(X1_train[i],X2_train[i])
}

sum(Y_train)
```

```
[1] 465
```

```r
training <- data.frame(X1_train=X1_train, X2_train=X2_train, Y_train=as.factor(Y_train))
ggplot(data=training, aes(x=X1_train, y=X2_train, color=Y_train)) +
  geom_point()
```



Fit team SVM

```r
svm_radial <- svm(Y_train ~ ., data=training, kernel="radial", gamma=0.1, cost=1)
summary(svm_radial)
```

```

Call:
svm(formula = Y_train ~ ., data = training, kernel = "radial", gamma = 0.1,
    cost = 1)


Parameters:
   SVM-Type:  C-classification
 SVM-Kernel:  radial
       cost:  1

Number of Support Vectors:  872

 ( 437 435 )


Number of Classes:  2

Levels:
 0 1
```

Test evaluation

Generate test

```r
# edit this
SEED <- 323

set.seed(SEED)

n = 100
X1_test <- runif(n,0,1)
X2_test <- runif(n,0,1)

Y_test <- rep(0,n)
```

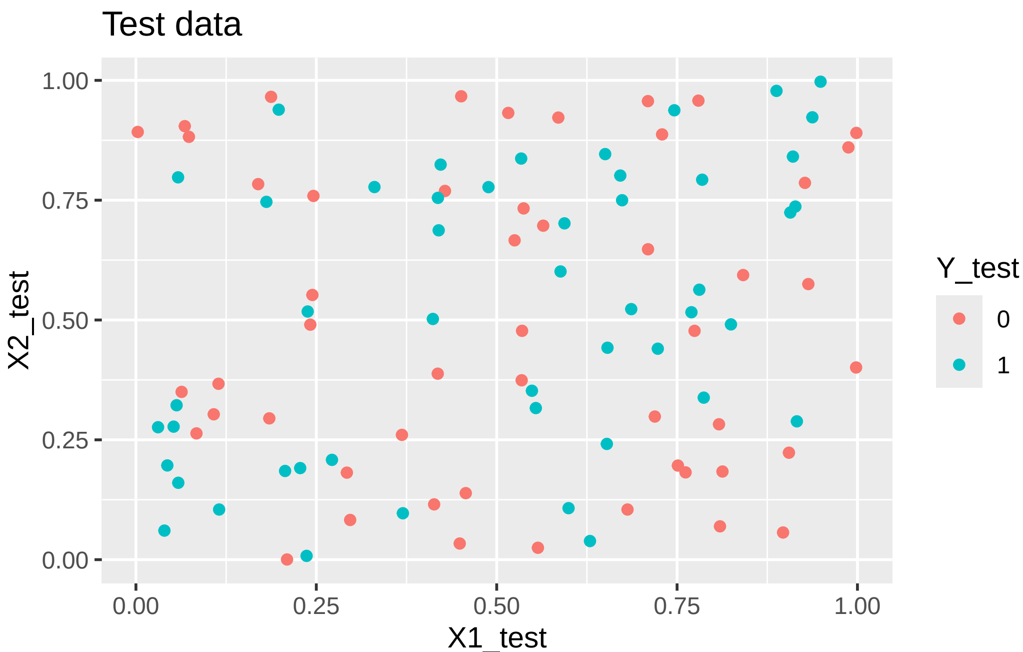
```r
for (i in 1:n) {
  Y_test[i] <- generate_y(X1_test[i],X2_test[i])
}

test <- data.frame(X1_test=X1_test, X2_test=X2_test, Y_test=as.factor(Y_test))

cat("# of 1s", sum(Y_test), "\n")
```

```
# of 1s 49
```

```r
cat("# of 0s", sum(1-Y_test), "\n")
```

```
# of 0s 51
```

```r
ggplot(data=test, aes(x=X1_test, y=X2_test, color=Y_test)) +
  geom_point() +
  ggtitle("Test data")
```



Evaluate team SVM

```r
# set the column names to match the training data
pred <- predict(svm_radial, newdata=data.frame(X1_train=X1_test, X2_train=X2_test))
sens <- sum(pred == 1 & Y_test == 1) / sum(Y_test == 1)
spec <- sum(pred == 0 & Y_test == 0) / sum(Y_test == 0)
misclass_error <- mean(pred != Y_test)
cat("Sensitivity:", sens, "\n")
```

```
Sensitivity: 0.3469388
```

```r
cat("Specificity:", spec, "\n")
```

```
Specificity: 0.7058824
```

```r
cat("Misclassification error:", misclass_error, "\n")
```

```
Misclassification error: 0.47
```